LLMs can’t jump

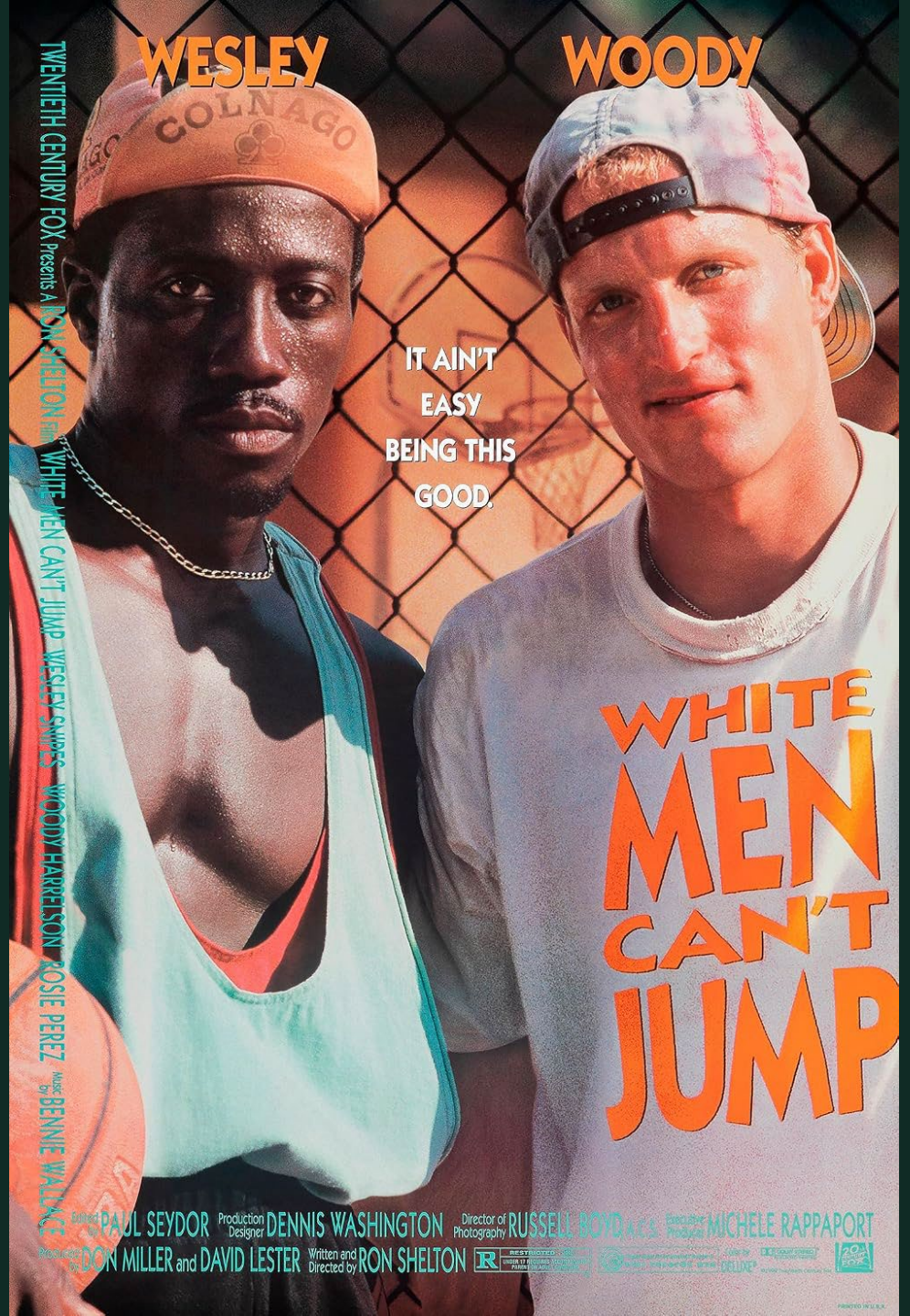


Tom Zahavy · Google DeepMind

ICML 2026 position paper

Theodore J. LaGrow

TReNDS Center MLBBQ · 10/23/26


WESLEY
WOODY
IT AIN'T EASY BEING THIS GOOD.
WHITE MEN CAN'T JUMP
TWENTIETH CENTURY FOX Presents A RON SHELTON Film WHITE MEN CAN'T JUMP WESLEY SNIPES WOODY HARRELSON ROSIE PEREZ
Edited by PAUL SEYDOR
Production Designer DENNIS WASHINGTON
Director of Photography RUSSELL BOYD A.C.S.
MICHELE RAPPAPORT
Produced by DON MILLER and DAVID LESTER
Written and Directed by RON SHELTON
R RESTRICTED


PLAY HARD. HUSTLE HARDER.
SINQUA WALLS
JACK HARLOW
20th CENTURY STUDIOS
WHITE MEN CAN'T JUMP
MAY 19
hulu
R
PERVASIVE LANGUAGE AND SOME DRUG MATERIAL



28

h-index

Google Scholar
23 Oct 2026

Tom Zahavy

Research Scientist

Google DeepMind, London

Co-leads the Discovery team

At DeepMind since 2019

PhD, Technion

Advisor: Shie Mannor

Who Is The Author?

Reinforcement learning before 2020

Early RL: DQN interpretability, hierarchical lifelong learning and action elimination. DeepMind: AlphaProof (RL proof search in Lean), creative chess and discovery.

Why this paper follows

His reasoning and creative-agent work motivates the question: can proof and search produce new scientific premises?

Two most-cited papers on his publication list

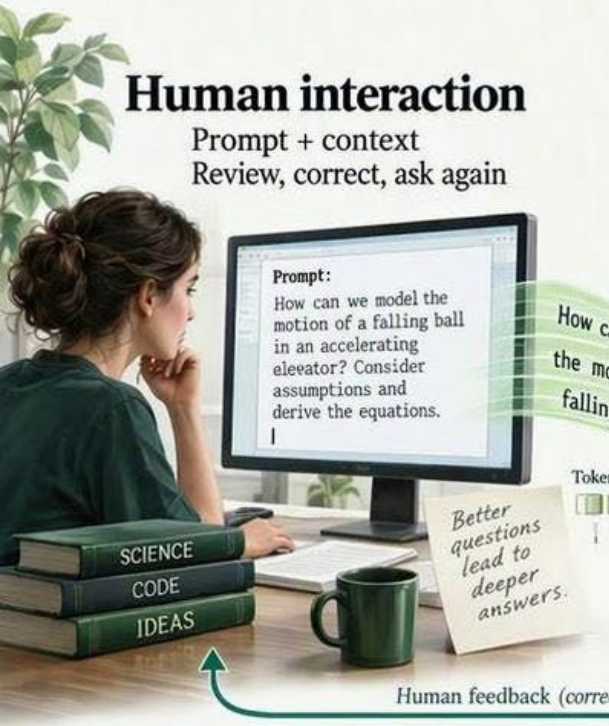
- A Deep Hierarchical Approach to Lifelong Learning in Minecraft (AAAI 2017)
- Graying the black box: Understanding DQNs (ICML 2016)

Inside an LLM reasoning system

Language prediction, human feedback and the frontier of scientific reasoning

Human interaction

Prompt + context
Review, correct, ask again



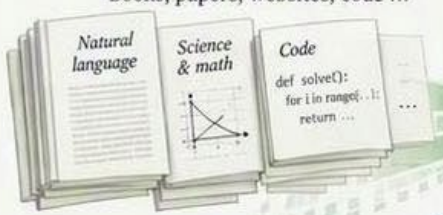
How can we model the motion of a falling ball ...

Token sequence

Training

Pretraining

Books, papers, websites, code ...

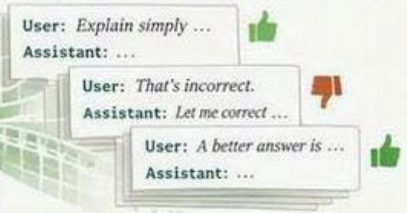


Learned parameters
(Billions of numbers)



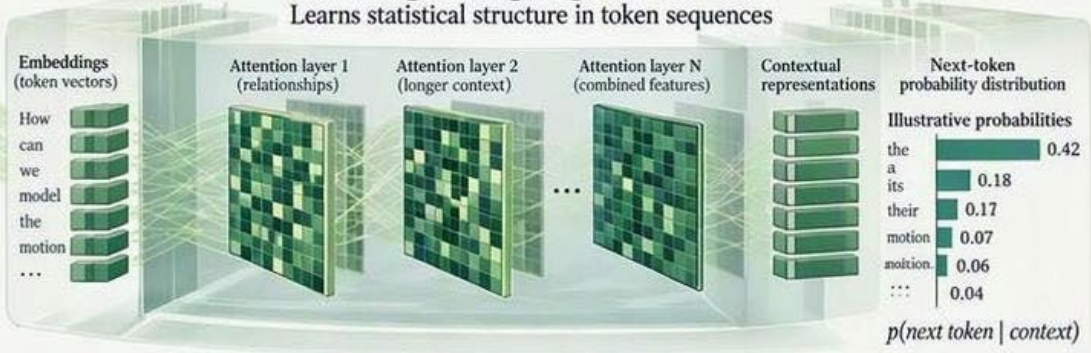
Instruction tuning / reinforcement learning

Human feedback, preferences, demonstrations



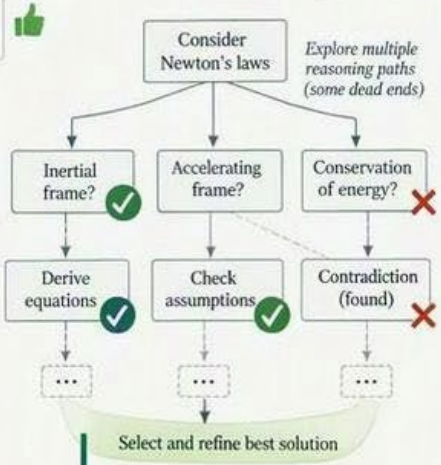
Large language model

Learns statistical structure in token sequences



Reasoning at inference

Intermediate steps, search and evaluation



Text, code or proposals

Answer:
Illustrative answer: In an elevator accelerating upward at a, a freely falling ball accelerates downward relative to the cabin at g + a (Newtonian approximation)."

Outputs still *require checking*

Human feedback *(corrections, new questions, preferences)*

AlphaProof

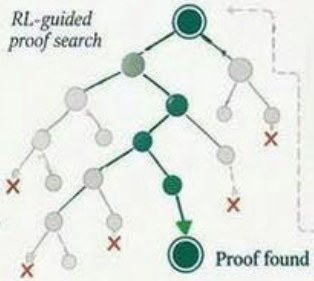
RL proof search with a formal verifier

Deduction from supplied premises

Mathematical statement
(e.g. IMO geometry problem)

Formal premises
Axioms, definitions, lemmas
...

RL-guided proof search



Formal proof (sketch)
1. From Axiom 1, ...
2. Therefore ...
3. Hence, ...
∴ Q.E.D.

Verified by formal checker

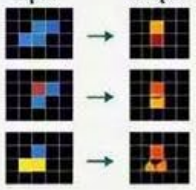
"A rigorous chain of reasoning can be learned and verified."

ARC-AGI

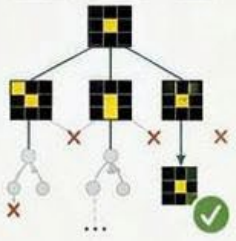
Sparse-example reasoning benchmark

Inference within supplied tasks

Example task
(3 input–output pairs)



Search for a consistent rule



Test on new input



Induces an abstract transformation that generalises to new cases.

Solve novel tasks from very few examples.

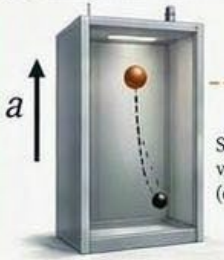
World-model grounding

Proposed extension in Zahavy's argument

PROPOSED EXTENSION
(not yet demonstrated)

Simulation to new axioms remains unresolved

Physical simulation
(e.g. falling ball in accelerating elevator)



Can simulation ground new axioms?

Simulate under varied conditions (discover regularities)

Candidate symbolic principles
1. In an accelerating frame, objects experience a fictitious force $F = -ma$.
2. Local physics is invariant ...
3. ...

?

How (and whether) such simulated regularities become new, reliable axioms remains an open question.

Conceptual synthesis, not one deployed architecture. Sources: Zahavy (2026), OpenAI (2024), Hubert et al. (2025).

World models: predicting the consequences of actions

A learned dynamics model supports imagined experiments and control

Observe

A camera observes the environment.

Adjustable gate (action a)

Encode state

Compress the observation into a latent state representation.

$$z(t+1) \sim p(z(t+1) \mid z(t), a(t))$$

z = latent state a = action

Visual observation $o(t)$

Encoder $q(z \mid o)$

$z(t)$ (latent state)

Predict under candidate actions

Imagine multiple futures from the current state.

a_1 Gate open

a_2 Gate partial

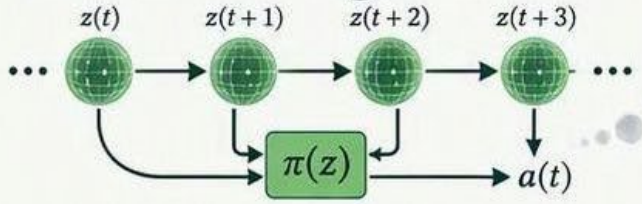
a_3 Gate closed

Predicted futures are uncertain. Errors can compound over time.

Choose action, act, compare with reality

Use the predicted outcome to select an action, perform it in the real world, and compare with the actual result to improve the model.

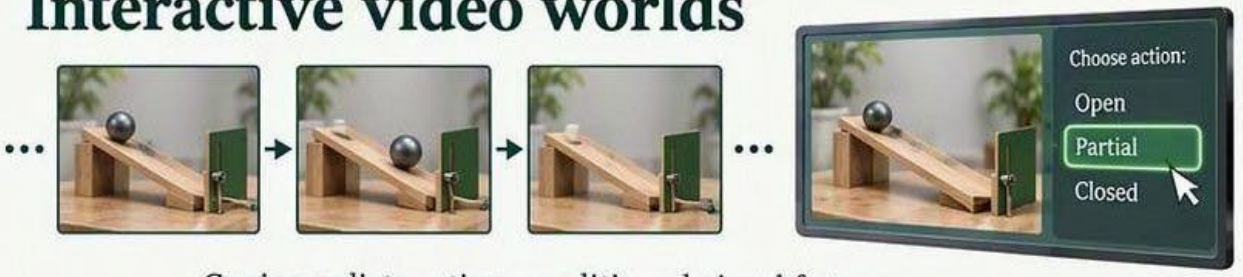
Latent imagination





DreamerV3 trains a policy on imagined trajectories. A learned policy can act without online rollout search.

Interactive video worlds



Genie predicts action-conditioned visual futures. Rendering pixels adds work beyond latent prediction.

Useful simulation does not by itself establish true physical laws or new axioms.

Conceptual illustration. References: Hafner et al., Nature (2025); Bruce et al., ICML (2024).

COMPUTE COMPARISON

Compute depends on the rollout budget

No universal “× LLM” cost. Compare the same task, quality target and hardware.

Approach	Compute per answer / decision	Additional preparation
Plain LLM	G(L, T) Prompt processing + token generation	Base-model training
RAG	C_ret + G(L + R, T) Retrieve, then read more context	Embed corpus + build index Optional retriever training
ReAct	Σ_i [G(L_i, T_i) + C_tool,i] Repeated model and tool calls	Can reuse the base LLM No new training required
World model with planning	C_enc + K H C_step + C_score Encode, simulate, evaluate	Learn dynamics from trajectories May also train a policy

ILLUSTRATIVE PLANNING BUDGET

1,024

64 candidate futures × 16 steps

= 1,024 model transitions

This is a transition count, not a 1,024× LLM cost. Each latent step may be much cheaper than an LLM call.

Video adds spatial tokens and frame decoding, sometimes iterative sampling.

Dreamer-style policy execution can avoid online rollouts entirely.

G = LLM compute, L/T = input/output tokens, R = retrieved tokens, K/H = rollout count/horizon. Analytical accounting, not a benchmark.

References: Lewis et al. (NeurIPS 2020), Yao et al. (ICLR 2023), Hafner et al. (Nature 2025), Bruce et al. (ICML 2024).

Where do scientific premises come from?

The author's central claim (verbatim)

> We argue that while Large Language Models have mastered the inductive compression of data and the deductive verification of theorems, they are structurally incapable of the abductive 'jump' required for scientific invention. We posit that this creative leap demands not just better language processing, but the integration of physically consistent World Models that ground abstract symbols in sensory simulation.

Induction

Explains regularities in observations.

Deduction

Derives consequences from supplied premises.

Abduction

Proposes explanations and new premises.

Zahavy's claim: grounded simulation supplies the missing step.

Counterfactual: could an AI restricted to Einstein's available knowledge invent general relativity?

The paper offers a historical argument and testable framework, not an impossibility theorem.

Source: Zahavy (2026), Position: LLMs can't jump, central claim.

Einstein's words in Zahavy's argument

"I see on one side the totality of sense experiences, and on the other, the totality of the concepts and propositions that are laid down in books. The relations between concepts and propositions among themselves are of a logical nature... The concepts and propositions get 'meaning', or 'content', only through their connection with sense experiences."

– *Albert Einstein*

"At the age of twelve I experienced a second wonder of totally different nature - in a little book dealing with Euclidean plan geometry, ..., were assertions, that could be proved with such certainty that any doubt appeared to be out of question. This lucidity and certainty made an indescribable impression on me."

– *Albert Einstein*

"Then there occurred to me the happiest thought of my life... for an observer falling freely from the roof of a house there exists—at least in his immediate surroundings—no gravitational field... The observer therefore has the right to interpret his state as 'at rest.' Because of this idea, the uncommonly peculiar experimental law that in the gravitational field all bodies fall with the same acceleration attained at once a deep physical meaning."

– *Albert Einstein*

"The words or the language, as they are written or spoken, do not seem to play any role in my mechanism of thought."

– *Albert Einstein*

Albert Einstein, as quoted in Zahavy (2026), Position: LLMs can't jump. Excerpts supplied by the presenter.

One elevator example, three inference directions

Illustrative toy rule: upward cabin acceleration makes a released object approach the floor.

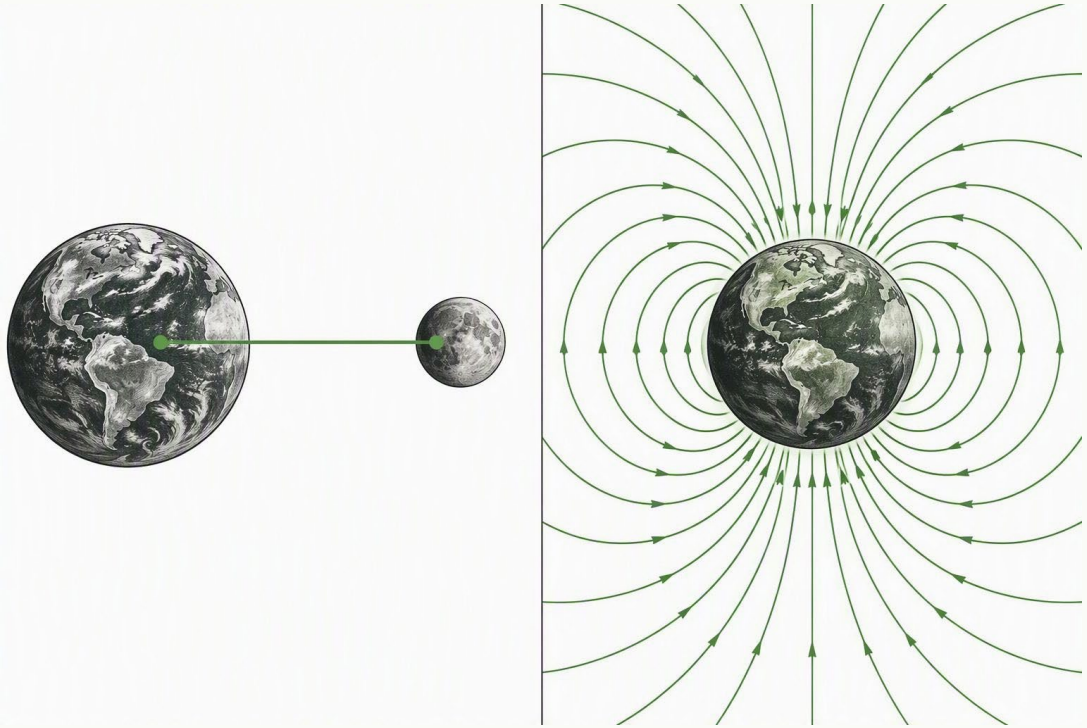
Deduction	Rule + accelerating cabin	→	**Predict apparent downward fall**
Induction	Cabin accelerations + observed falls	→	**Infer a general relation**
Abduction	Rule + apparent downward fall	→	**Hypothesize cabin acceleration**

Abduction proposes a possible cause, not a unique conclusion: gravity could also explain the fall.

The stronger target is inventing a new physical premise, rather than selecting a known explanation.

Context: Bhagavatula et al., Abductive Commonsense Reasoning (ICLR 2020).

Good predictions left a conceptual tension



Empirical success

Newtonian gravity fit much of the evidence. Mercury's unexplained orbital precession could motivate a patch such as the proposed planet Vulcan.

Conceptual conflict

Point masses acted across distance, while field theories described continuous local structure. Zahavy argues that resolving this tension required new premises.

Why it matters: a low prediction error need not settle which concepts a theory should use.

Historical context: Norton, Einstein's Pathway to General Relativity (2024 archive).

Special relativity supplies the starting premises

Relativity principle

The same physical laws hold in every inertial frame.

Light-speed invariance

Every inertial observer measures the same vacuum speed c.

$$t' = \gamma(t - vx/c^2)$$

$$\gamma = 1 / \sqrt{(1 - v^2/c^2)}$$

v: relative frame velocity x, t: original coordinates t’: transformed time

Interpretation: simultaneity depends on the frame because transformed time also depends on position x.

Next challenge: incorporate gravity and accelerated observers into the physical framework.

The historical search mixed intuition and formal work

1907–1912

Ideation

Equivalence insight links free fall and gravity.

Static-field models explore consequences for clocks and light.

1912–1913

Consolidation

Grossmann helps express gravity through curvature.

The Entwurf theory offers a candidate mathematical framework.

1913–1915

Correction

Consistency tests force revision.

Nov. 18, 1915: Mercury result.

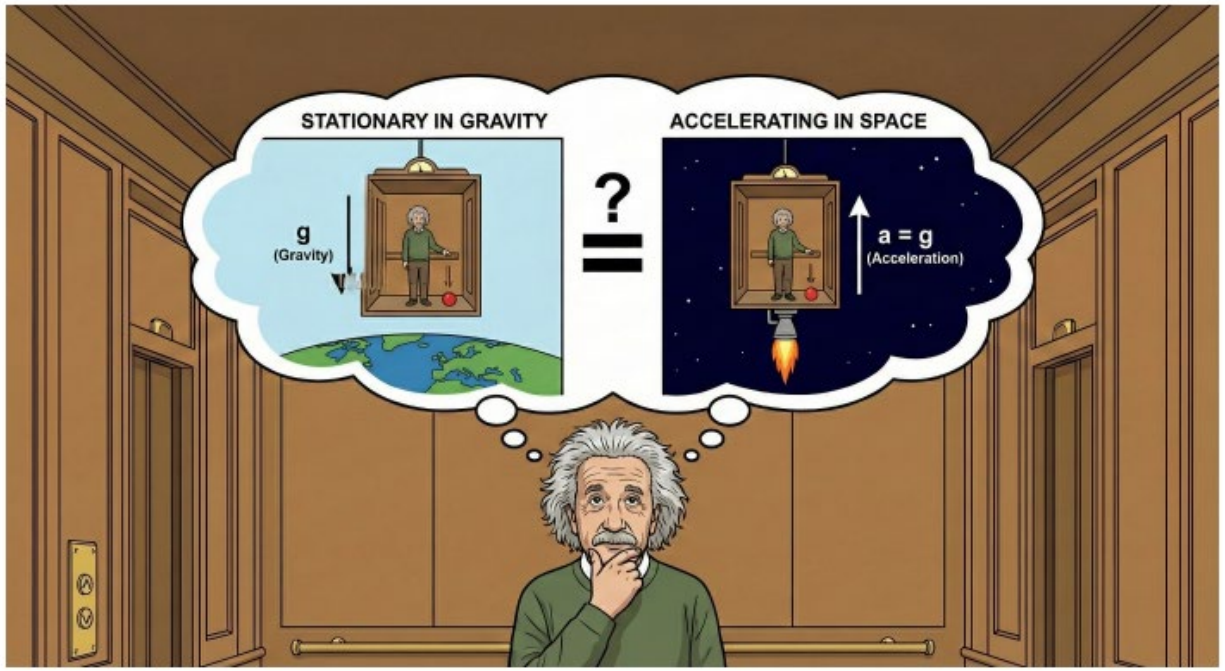
Nov. 25: final field equations.

The phases describe a historical trajectory, not cleanly separated cognitive modules.

Historical context: Norton, Einstein's Pathway to General Relativity (2024 archive).

The elevator makes local equivalence concrete

Figure 2 from the paper



Released test body

In empty space, the accelerating cabin floor catches a released body. Locally, this mimics a body falling in a gravitational field.

$$m_i\, a = m_g\, g$$

$m_i = m_g$ gives $a = g$

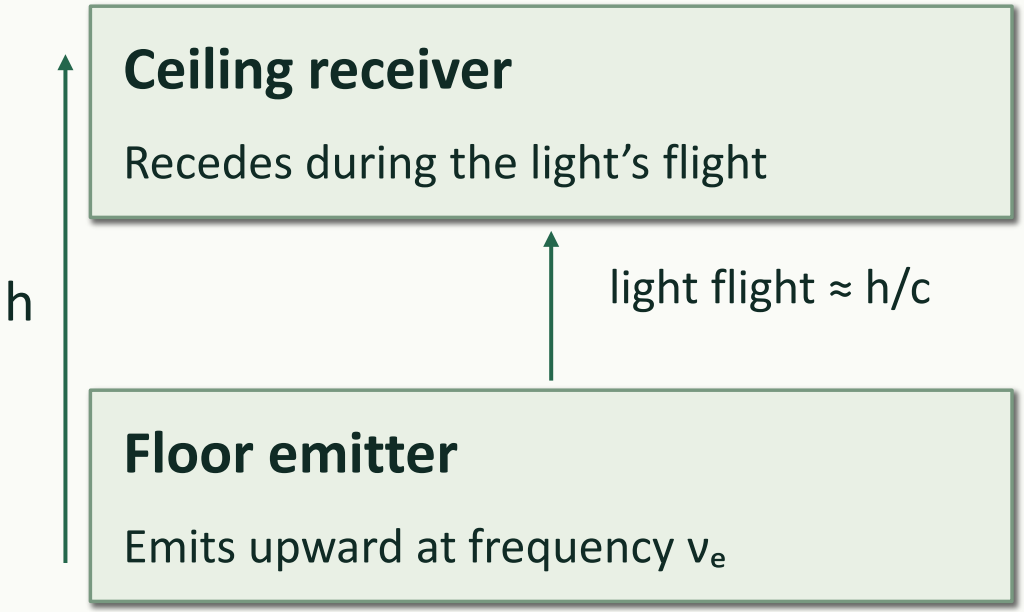
m_i: inertial mass mg: gravitational mass

Local comparison only. Tidal effects can distinguish extended gravitational fields.

Zahavy (2026), Fig. 2. AI-generated thought-experiment illustration in the source paper.

A worked consequence: light loses frequency uphill

Accelerating cabin, height h



Weak-field, first-order argument

$$\Delta t \approx h/c$$

$$\Delta v \approx a\Delta t = ah/c$$

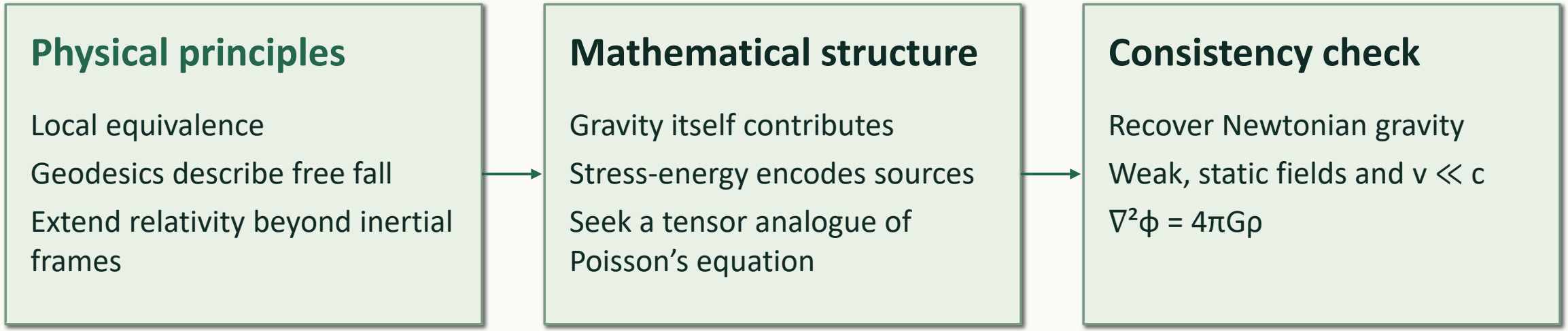
$$(\nu_r - \nu_e)/\nu_e \approx -\Delta v/c$$

$$a = g \Rightarrow \Delta\nu/\nu_e \approx -gh/c^2$$

Weak field: $gh/c^2 \ll 1$. Evidence: Bothwell et al., Nature (2022).

Measured clock-rate variation across a millimetre-scale atomic sample.

Physical premises become equations and checks



ϕ: gravitational potential ρ: mass density G: Newton's constant.

These are constraints, not a complete derivation.

1913: a mistaken static-field assumption made promising geometry appear to fail the Newtonian limit.

A failed consistency check may indict the auxiliary assumption, not the whole theory.

Historical context: Norton, Einstein's Pathway to General Relativity (2024 archive).

The field equation couples curvature to matter

$$R_{ik} - \tfrac{1}{2} g_{ik} R = -\kappa T_{ik}$$

The left side is the Einstein tensor. The equation constrains geometry and its matter–energy sources.

Spacetime geometry

g_{ik} metric: distances and time intervals

R_{ik} Ricci curvature

R scalar curvature

Matter and energy

T_{ik} energy, momentum and stresses

κ coupling constant

i, k spacetime indices

Components of T encode energy density, energy flux, pressure and shear stress, with convention-dependent units.

Paper Eq. (2). Sign and normalization follow the author's conventions.

Discovery closes the loop between ideas and experience

Figure from Zahavy

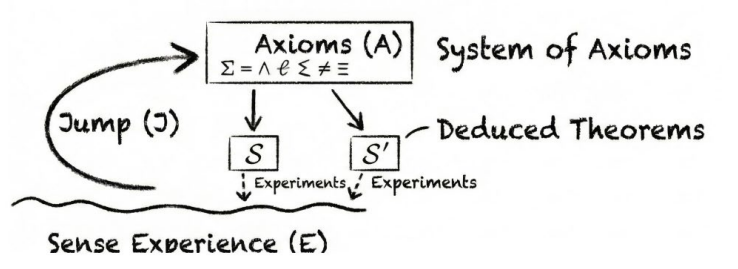
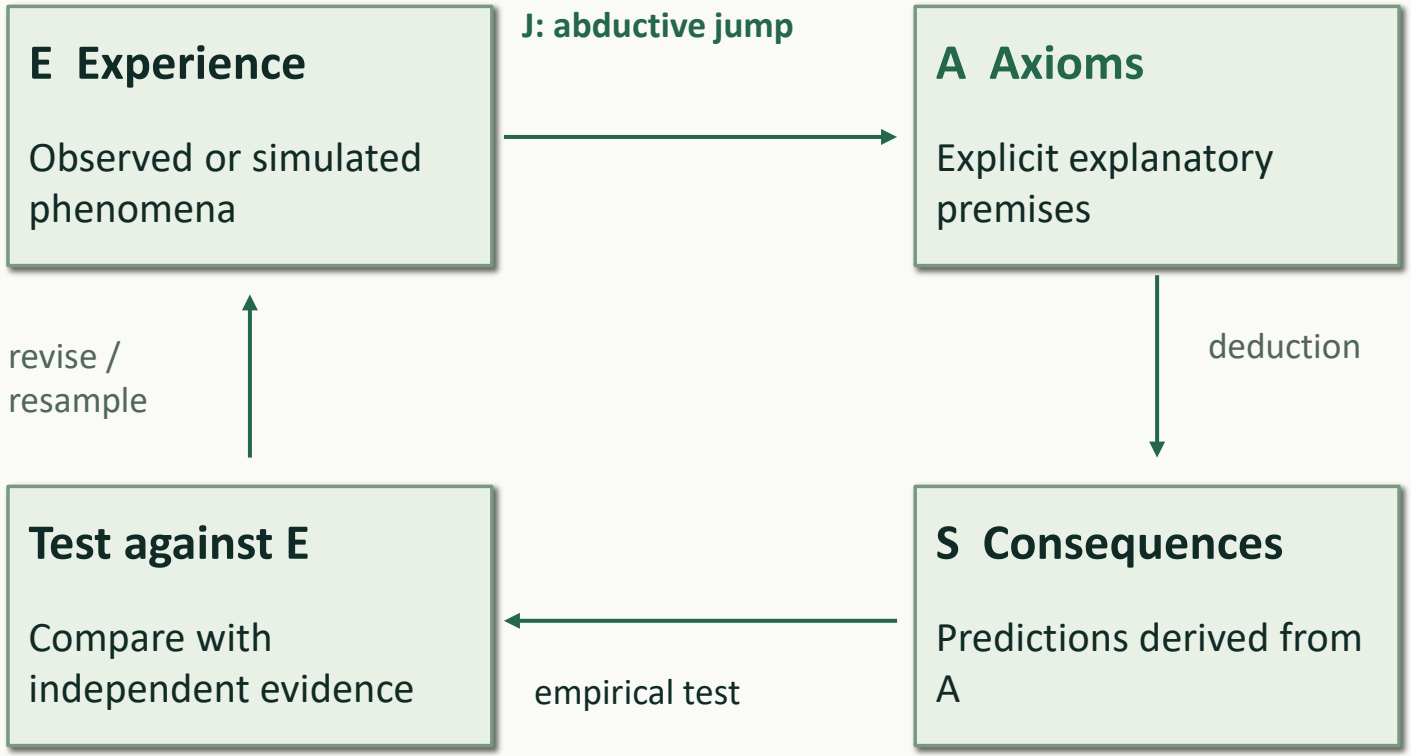


Figure 1, p. 1

The paper's AI-generated reconstruction of Einstein's diagram.

Explanatory reconstruction



The disputed capability is proposing grounded premises at J.

Source: Zahavy (2026), Position: LLMs can't jump, Figure 1. Original figure reproduced without redrawing.

Equation discovery is real, but representation matters

Illustrative MDL objective: $L(H) = \text{error}(H, D) + \lambda\ \text{complexity}(H)$

H: candidate theory D: observations λ: trade-off between fit and complexity

Demonstrated

SINDy selects a sparse set of terms from a candidate function library to fit measured dynamics.

AI Feynman uses physics-inspired structure to recover symbolic equations from supplied data.

Boundary of the evidence

The supplied variables, function library and structural priors shape what can be found.

Recovering an equation differs from introducing a new explanatory concept or physical ontology.

Open question: can a system revise its explanatory vocabulary?

Brunton, Proctor & Kutz, PNAS (2016). Udrescu & Tegmark, Science Advances (2020).

AlphaProof: learned search for verifiable proofs

The result: IMO 2024

3 of 5

Non-geometry problems solved at the International Mathematical Olympiad, including its hardest problem.

The full contest had six problems. AlphaGeometry 2 solved the geometry problem, bringing the combined result to 4/6 (28/42 points), a silver-medal standard.

How AlphaProof works

1 Learn proof search

An AlphaZero-inspired agent uses RL to learn from millions of auto-formalized problems.

2 Adapt to a difficult problem

Test-time RL learns from related problem variants, adapting the model rather than only sampling more answers.

3 Check the proof

Lean checks each formal proof against the encoded statement and assumptions.

A ⊢ S: encoded premises A formally entail conclusion S.

Why it matters here: AlphaProof demonstrates powerful deduction. It does not establish that the premises describe nature or show how to invent new physical axioms.

Sources: Hubert et al., Nature (2025); Google DeepMind, IMO announcement (2024). Zahavy is a coauthor.

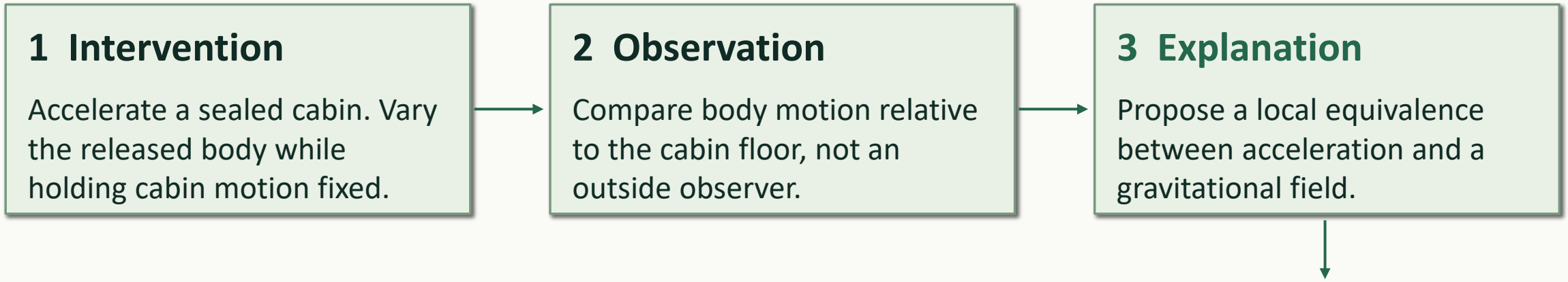
Sparse examples leave the experience supplied

Task	Given to the system	What remains to infer
ARC-style task	2–5 example grid pairs (the paper's description)	Infer a transformation and apply it to a new grid
Symbolic variation	A formal system or game with fixed rules	Explore new cases or strategies within those rules
Physical thought experiment	Physical priors and an intervention the agent can vary	Observe a consequence and propose an explanatory premise

Zahavy's distinction concerns who creates the informative experience, not just how many examples are shown.

The author's extra requirement: invent an experiment that makes a new concept available.

The elevator example makes the jump explicit



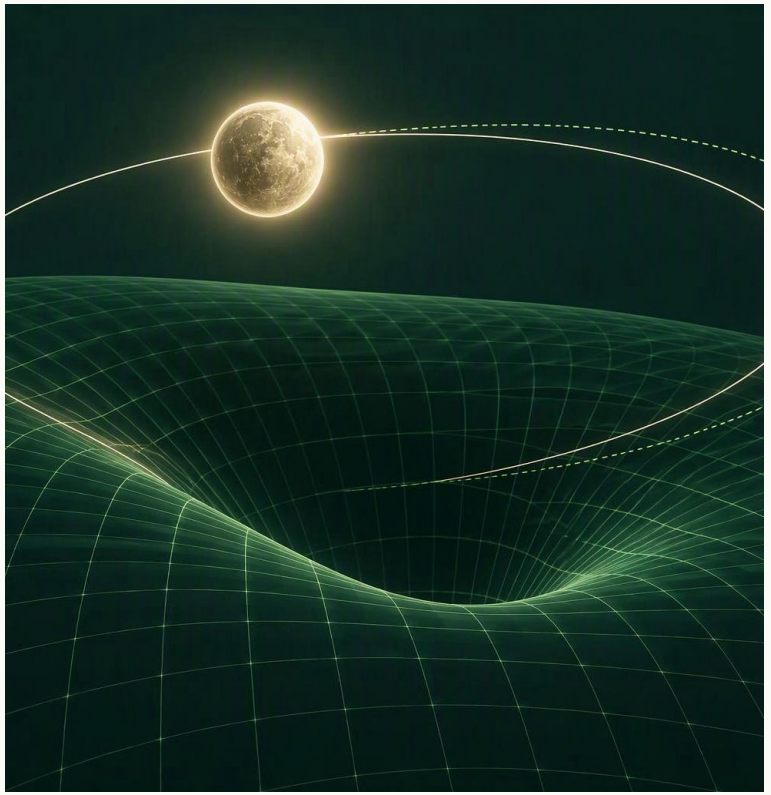
Formal premise A

A homogeneous gravitational field and a uniformly accelerating frame give the same local experimental outcomes.

Engineering gap: translate a perceptual pattern into explicit premises with variables, scope and assumptions.

This is hypothesis formation. The observation alone does not prove the premise.

Controllability and physical understanding differ



Genie

Learns latent actions from unlabeled video and supports frame-by-frame control.

Control demonstrates a usable action interface. It does not certify physical laws or reliable interventions outside the training distribution.

Inductive-bias probes

Vafa et al. probe whether learned orbital predictions transfer as a Newtonian model to new tasks. Good in-task prediction can coexist with poor structural transfer.

Scope: evidence about the evaluated models and probes, not all possible world models.

Bruce et al., Genie, ICML (2024). Vafa et al., What Has a Foundation Model Found?, ICML (2025).

Our proposed test: an explicit discovery algorithm

Inputs: W simulator, P prior, V verifier, search budget B

```
repeat until budget B is exhausted:
  E ← simulate(W, choose_intervention(P))
  H ← propose_explanation(E, P)
  A ← translate_to_axioms(H)       # unresolved
  S ← deduce(A)
  if V(A, S) fails: revise A and continue
  if validation(S) fails: revise W, P, or A
  else: freeze candidate and end search
if no candidate: return failure
evaluate once on blind test worlds; report all outcomes
```

Validation guides revision. Blind final tests never feed back into this search.

Presenter proposal based on Zahavy §4. No implemented axiom translator or reported results.

The proposed test separates agency from modality

Vary one factor at a time

Modality

Same information as text or visual observations

Agency

Chosen actions vs replay of identical observations

Representation

Free-form explanation vs explicit, testable axioms

Separate validation worlds from the final blind test.

Blind-test success

Freeze the hypothesis and system before testing. Predict unseen intervention outcomes within a predeclared tolerance.

Match data, compute, tools and priors.

Report failures and independent-world transfer.

Score formal validity separately from predictive accuracy.

Motivation: ScienceWorld, Wang et al., EMNLP (2022), uses an interactive text environment.

Demonstrated capabilities and their remaining gaps

Capability and study	Demonstrated result	Remaining gap
Equation recovery SINDy / AI Feynman	Equations from data with structural priors	New physical vocabulary beyond supplied representations
Formal proof AlphaProof	Lean proofs using RL and verification	Autonomous generation of physical premises
Novel program search FunSearch	New constructions via LLM + evaluator	New physical ontology from text alone
Interactive simulation Genie	Action-controllable learned environments	Physical fidelity and valid new axioms

These studies demonstrate different capabilities. None settles the full invention claim.

Brunton et al. 2016; Udrescu & Tegmark 2020; Hubert et al. 2025; Romera-Paredes et al. 2023; Bruce et al. 2024.

Novel outputs already exist within specified tasks



FunSearch and AlphaEvolve

FunSearch found new cap-set constructions and bin-packing heuristics through LLM code proposals plus evaluation.

AlphaEvolve extends evaluator-guided code search. Its 2025 white paper reports results within specified problems.

LLM hypothesis generation

Zhou et al. generate hypotheses from labeled examples that improve prediction in the studied classification tasks.

Novel programs or task hypotheses differ from new physical axioms.

Romera-Paredes et al., Nature (2023 online); Novikov et al., white paper (2025); Zhou et al., NLP4Science (2024).

A useful question. An unproven mechanism.

Evidence

Formal proofs, equation recovery, novel programs and controllable simulations each demonstrate parts of discovery.

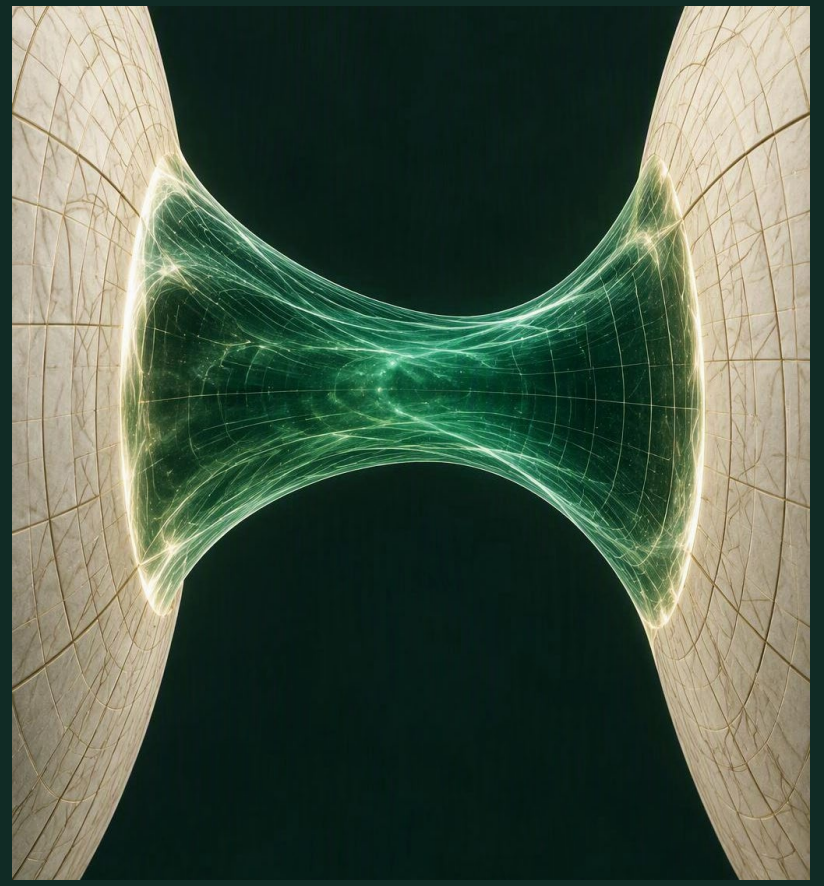
Unproven claim

Necessity: must discovery have sensory grounding?

Sufficiency: can a world model generate valid new physical premises?

Testable next step

Match information and compute. Separate agency from modality. Freeze systems before blind-world evaluation.



Zahavy (2026), Position: LLMs can’t jump. Core argument: §§3–5 and Appendix A.

Limitations and criticism of the paper

Presenter assessment: a provocative research agenda with an untested causal explanation.

1 Strong claim, limited evidence

“Structurally incapable” exceeds the evidence: the paper offers no impossibility proof or controlled comparison of discovery systems.

2 History cannot isolate the cause

Einstein’s pathway illustrates discovery. It cannot establish that sensory grounding is necessary, or that formal and linguistic routes must fail.

3 The proposed bridge is missing

The simulation-to-axiom step remains unspecified. Learned worlds may reproduce their priors or errors rather than reveal new physical laws.

4 Novelty needs a precise test

FunSearch shows novelty within defined tasks. The stronger claim needs a test that distinguishes new physical premises from rediscovery.

Needed evidence: matched baselines, explicit novelty criteria and independent-world tests of frozen systems.

Basis: Zahavy (2026), §§3–5; Romera-Paredes et al., Nature (2023 online/2024); Vafa et al., ICML (2025).

What the paper does not yet establish

Presenter assessment: plausible motivation does not identify the mechanism of discovery.

Confounded explanations

Any gain could reflect better priors, more information or more search. A fair test must isolate interaction from these alternatives.

Simulator circularity

If the simulator contains the target physics, recovered “laws” may expose its built-in assumptions. Independent-world transfer is essential.

Unclear novelty criteria

Without an operational definition of a new premise, success can be labeled recombination and failure blamed on implementation. This weakens falsifiability.

An untested mechanism

The simulation-to-axiom mechanism is unspecified. Necessity of grounding and sufficiency of world models require separate evidence.

Bottom line: a valuable research agenda, but the causal and architectural conclusions remain open.

Basis: Zahavy (2026), §§3–5; methodological implications of the evidence and proposed tests in this deck.

What result would change your view?

01 **Novelty**

What distinguishes a new axiom from recombination?

Candidate criterion: a new explanatory vocabulary with testable consequences.

02 **Grounding**

Does intervention help when information is matched?

Control: compare chosen interventions with a replay of identical observations.

03 **Falsification**

What would refute the claim without training-data leakage?

Test: unfamiliar worlds, frozen systems and a success threshold chosen in advance.

DISCUSSION

1 Novelty and the claim

What counts as a new axiom rather than recombination? How would we distinguish a structural limit from a resource limit?

2 Grounding and causality

Does intervention help when information is matched? How can we separate discovered laws from simulator assumptions?

3 Evidence and falsification

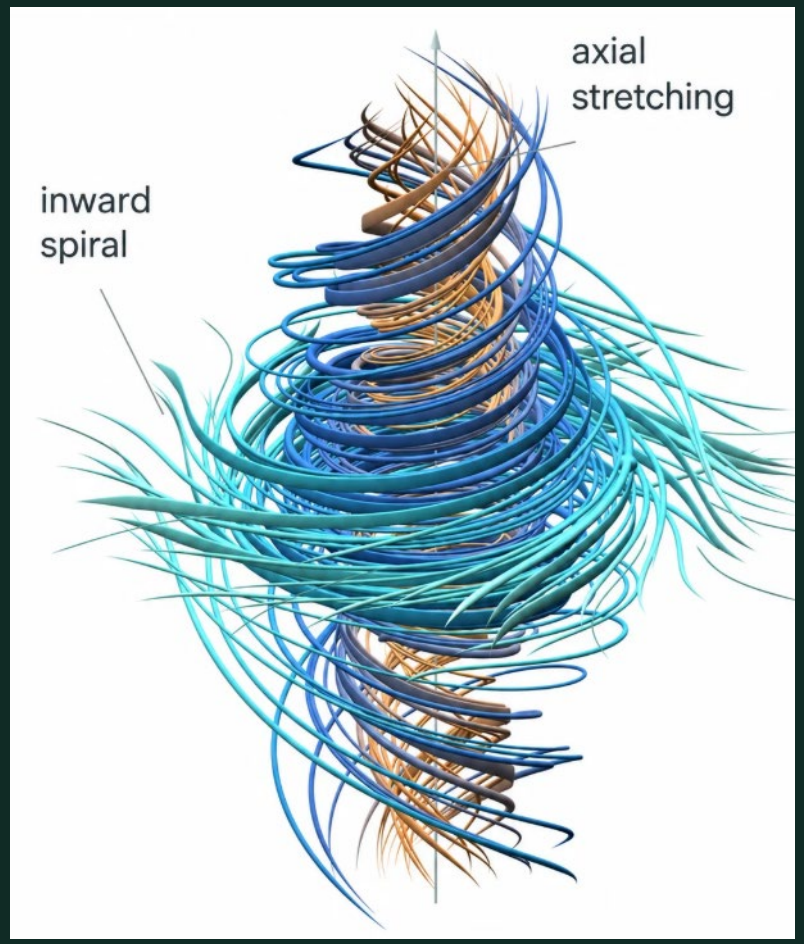
Which blind test, matched baselines and independent metrics would distinguish genuine invention from rediscovery?

4 Compute and a decisive experiment

At fixed compute, when do world models beat LLM, RAG or ReAct search? What minimal experiment could change your view?

Choose one: state a hypothesis, its strongest rival and an outcome that would distinguish them.

OpenAI’s Navier–Stokes claim



Inward spiraling and axial stretching

User-supplied illustration of the proposed vortex

The Millennium Prize problem

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \quad \nabla \cdot u = 0$$

In 3D, can smooth incompressible flow develop a finite-time singularity? Here u is velocity, p pressure, $\nu > 0$ viscosity and f forcing.

What OpenAI reports (8 September 2026)

An internal AI system produced a blowup proof and Lean formalization. The reported construction uses smooth forcing, starts from rest and has unbounded velocity despite bounded kinetic energy.

What would this establish?

A formal theorem within known equations tests mathematical discovery. Does it also test the invention of new physical axioms?

Reported result, not a claim here of independent acceptance. Sources: OpenAI (8 Sep 2026); Fefferman, Clay problem statement.

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